Full calculation of QFI for the SW1 approximated Hamiltonian

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Appendix to Critical Metrology

From the s_1 and z obtained previously an approximate ground state vector to the QRM is

$$|GS_{\text{QRM}}\rangle = e^{-\frac{ig}{2\Delta}(a^\dagger+a)\sigma_y}\, e^{-\frac{1}{8}\ln\left(1-\frac{2g^2}{\omega\Delta}\right)(a^{\dagger 2}-a^2)}\, |0_a\rangle \otimes |\downarrow\rangle\,. \tag{1}$$

It is a great sanity check to check that for $g = 0$ the ground state is $|GS_{QRM}\rangle = |0\downarrow\rangle$, which is obvious.

We are ready to compute the QFI by applying the general formula for a pure state. It is going to be painfully tedious, so prepare yourself. We start by computing $\frac{\partial}{\partial g}|GS_{\text{QRM}}\rangle$

$$\frac{\partial}{\partial g}|GS_{\text{QRM}}\rangle = -\frac{i}{2\Delta}(a^\dagger + a)\sigma_y e^{-s_1} S(r)|0\rangle \otimes |\downarrow\rangle + e^{-s_1}\frac{1}{2}(a^{\dagger 2} - a^2)S(r)\frac{\partial r}{\partial g}|0\rangle \otimes |\downarrow\rangle\,.$$

Using that $\langle GS_{\mathbf{QRM}}| = \langle 0| \otimes \langle\downarrow| S(-r)e^{s_1}$[1] we compute $\langle GS_{\mathbf{QRM}}|\partial_g GS_{\mathbf{QRM}}\rangle$

$$\langle GS_{\text{QRM}}|\partial_g GS_{\text{QRM}}\rangle = \langle 0|\otimes\langle\downarrow|\; S(-r)\, e^{s_1}\left[-\frac{i}{2\Delta}(a^\dagger + a)\,\sigma_y\, e^{-s_1} S(r) + \frac{1}{2}\, e^{-s_1}(a^{\dagger 2} - a^2)S(r)\,\frac{\partial r}{\partial g}\right]|0\rangle\otimes|\downarrow\rangle\;.$$

This horrible and very intimidating equation is not as threatening as it looks. We have been blessed, because $[s_1, \sigma_y] = [s_1, (a + a^\dagger)] = 0$, which means that $e^{s_1}\sigma_y(a + a^\dagger)e^{-s_1} = \sigma_y(a + a^\dagger)e^{s_1}e^{-s_1} = \sigma_y(a + a^\dagger)$ and $S(r)$ does not act on σ_y.

This enables us to write the first term as

$$-\frac{i}{2\Delta}\;\langle 0|\; S(-r)\,(a^\dagger + a)\,S(r)\,|0\rangle\,\langle\downarrow|\,\sigma_y\,|\downarrow\rangle = 0.$$

For the second term we have

$$\frac{\partial_g r}{2}\,\langle 0\downarrow|S(-r)e^{s_1}e^{-s_1}(a^{\dagger 2} - a^2)S(r)|0\downarrow\rangle\,,$$

here $e^{s_1}e^{-s_1} = 1$, simplifying the calculations. Since $(a^{\dagger 2} - a^2)$ is the generator of squeezing, it is left invariant under the action of the squeezing operator (can be seen via BCH). We are left with

$$\frac{\partial_g r}{2}\,\langle 0\downarrow|(a^{\dagger 2} - a^2)|0\downarrow\rangle = 0,$$

So, we have found that $\langle GS_{\text{QRM}}|\partial_g GS_{\text{QRM}}\rangle = 0$, as both terms are zero.

Now, we need to compute the overlap of the derivatives.

For what comes next, we need to have the following expressions at our disposal

$$|\partial_g GS_{\text{QRM}}\rangle = -\frac{i}{2\Delta}\,(a^\dagger + a)\,\sigma_y\, e^{-s_1} S(r)\;|0\rangle \otimes |\downarrow\rangle\;+\;\frac{1}{2}\,(\partial_g r)\, e^{-s_1}(a^{\dagger 2} - a^2)S(r)\;|0\rangle \otimes |\downarrow\rangle\,,$$

$$\langle\partial_g GS_{\text{QRM}}| = \frac{i}{2\Delta}\;\langle 0|\otimes\langle\downarrow|\; S(-r)\, e^{s_1}\,\sigma_y\,(a + a^\dagger)\;+\;\frac{1}{2}\,(\partial_g r)\;\langle 0|\otimes\langle\downarrow|\; S(-r)\,(a^2 - a^{\dagger 2})\, e^{s_1},$$

[1] $S^\dagger(r) = S(-r)$

We will write this derivatives using A and B, to make the overlap simpler

$$|A\rangle = -\frac{i}{2\Delta}\left(a^\dagger + a\right)\sigma_y e^{-s_1} S(r)\,|0\rangle \otimes |\downarrow\rangle\,,$$
$$|B\rangle = \frac{1}{2}(\partial_g r)e^{-s_1}\left(a^{\dagger 2} - a^2\right) S(r)\,|0\rangle \otimes |\downarrow\rangle\,.$$

We are lucky enough to have $\langle A|B\rangle = \langle B|A\rangle = 0$, due to the appearance of a σ_y in both cases. So,

$$\langle \partial_g GS_{\text{QRM}}|\partial_g GS_{\text{QRM}}\rangle = \langle A|A\rangle + \langle B|B\rangle\,.$$

Term $\langle A|A\rangle$:

Using $[\sigma_y, s_1] = [(a + a^\dagger), s_1] = 0$ and $\sigma_y^2 = 1$, we get:

$$\langle A|A\rangle = \frac{1}{4\Delta^2}\langle 0|\,S(-r)(a + a^\dagger)^2 S(r)\,|0\rangle\,.$$

In the squeezed vacuum, $S^\dagger(r)(a + a^\dagger)^2 S(r) = e^{-2r}(a + a^\dagger)^2$ (refer to **??**), hence:

$$\langle 0|\,S(-r)(a + a^\dagger)^2 S(r)\,|0\rangle = e^{-2r}.$$

Therefore,

$$\langle A|A\rangle = \frac{e^{-2r}}{4\Delta^2}.$$

Term $\langle B|B\rangle$:

Here e^{s_1} and e^{-s_1} cancel adjacently, and since $S(-r)(a^{\dagger 2} - a^2)S(r) = (a^{\dagger 2} - a^2)$ (it's the generator of squeezing),

$$\langle B|B\rangle = \frac{(\partial_g r)^2}{4}\langle 0|(a^{\dagger 2} - a^2)(a^2 - a^{\dagger 2})|0\rangle\,.$$

Using $a^{\dagger 2}a^2 = a^{\dagger 2}a^2 + 4a^\dagger a + 2$ and $\langle 0|a^\dagger a|0\rangle = 0$,

$$\langle 0|(a^{\dagger 2} - a^2)(a^2 - a^{\dagger 2})|0\rangle = 2,$$

yielding the final result

$$\langle B|B\rangle = \frac{(\partial_g r)^2}{2}.$$

Result:

We FINALLY have

$$\langle \partial_g GS_{\text{QRM}}|\partial_g GS_{\text{QRM}}\rangle = \frac{e^{-2r}}{4\Delta^2} + \frac{1}{2}(\partial_g r)^2$$

resulting in the following form for the QFI

$$F_Q(g) = 4\langle \partial_g\psi|\partial_g\psi\rangle = \frac{e^{-2r}}{\Delta^2} + 2(\partial_g r)^2$$

Adopting the $r(g)$ we found before

$$r(g) = -\frac{1}{4}\ln\left(1 - \frac{2g^2}{\omega\Delta}\right), \quad \partial_g r = \frac{g}{\omega\Delta - 2g^2}, \quad e^{-2r} = \sqrt{1 - \frac{2g^2}{\omega\Delta}},$$

then

$$F_Q(g) = \frac{\sqrt{1 - \frac{2g^2}{\omega\Delta}}}{\Delta^2} + \frac{2g^2}{(\omega\Delta - 2g^2)^2}. \tag{2}$$